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Paddy Rice Mapping in Hainan Island Using Time-Series Sentinel-1 SAR Data and Deep Learning

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Abstract: Rice serves as a fundamental staple food for a significant portion of the global population, and accurate monitoring of paddy rice cultivation is essential for achieving Sustainable Development Goal (SDG) 2–Zero Hunger. This study proposed two models, RiceLSTM and RiceTS, designed for the precise extraction of paddy rice areas in Hainan Island using time-series Synthetic Aperture Radar (SAR) data. The RiceLSTM model leverages a Bidirectional Long Short-Term Memory (BiLSTM) network to capture temporal variations in SAR backscatter and integrates an attention mechanism to enhance sensitivity to paddy rice phenological changes. This model achieves classification accuracies of 0.9182 and 0.9245 for early and late paddy rice, respectively. The RiceTS model extends RiceLSTM by incorporating a U-Net architecture with MobileNetV2 as its backbone, further improving the classification performance, with accuracies of 0.9656 and 0.9808 for early and late paddy rice, respectively. This enhancement highlights the model's capability to effectively integrate both spatial and temporal features, leading to more precise paddy rice mapping. To assess the model's generalizability, the RiceTS model was applied to map paddy rice distributions for the years 2020 and 2023. The results demonstrate strong spatial and temporal transferability, confirming the model's adaptability across varying environmental conditions. Additionally, the extracted rice distribution patterns exhibit high consistency with statistical data, further validating the model's effectiveness in accurately delineating paddy rice areas. This study provides a robust and reliable approach for paddy rice mapping, particularly in regions that are characterized by frequent cloud cover and heavy rainfall, where optical remote sensing is often limited.

Keywords: paddy rice; mapping; SAR; Sentinel-1; temporal–spatial; deep learning; RiceLSTM; RiceTS

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1. Introduction

Ensuring a sufficient food supply is a fundamental objective of the United Nations' 2030 Agenda for Sustainable Development Goals (SDGs). As a major crop, rice plays a key role in global food security, serving as the primary staple food for more than 50% of the world's population [1], particularly in Asia [2]. Over 90% of the world's rice is produced in the Asia–Pacific Region, with major production centers in East Asia, South Asia, and Southeast Asia [3]. However, rapid urbanization and industrialization introduce uncertainty in paddy rice cultivation, alongside other influential factors such as weather conditions, water availability, paddy rice diseases and pests, agricultural practices, and

economic policies [4]. Concurrently, the International Food Policy Research Institute (IFPRI) reports that the rice demand is growing at an annual rate of 1.8% [5]. Given these challenges, the timely and accurate monitoring of paddy rice planting areas, growth dynamics, and yield estimation are essential for informed agricultural policymaking and the realization of SDG 2–Zero Hunger.

The traditional methods for large-scale paddy rice monitoring are time-consuming and labor-intensive. Remote sensing technology has become a key tool for regional and global crop mapping, offering broad spatial coverage, frequent data acquisition, and reduced observation costs. It provides precise and timely insights into crop phenology and development [3,6]. Several studies have been conducted on the use of coarser- to finer-resolution optical remote data for crop mapping [7] and paddy rice mapping [8,9], mainly including MODIS, Landsat, and Sentinel-2 data. Time-series Landsat-8 OLI data, coupled with a deep learning (DL) model, FR-Net, successfully mapped paddy rice in the main rice-producing area of northern China, demonstrating the significant potential of Landsat data in agricultural remote sensing [10]. Similarly, 30 m harmonized Landsat and Sentinel-2 surface reflectance data, processed using the Feature Selection and Hierarchical Classification (FSHC) method, were applied for paddy rice mapping in South China [11].

However, optical remote sensing faces challenges in paddy rice regions, which are frequently cloudy and rainy, limiting effective data acquisition during critical growth stages such as transplanting and flooding. Synthetic Aperture Radar (SAR) overcomes these limitations by offering all-weather, day-and-night imaging [12]. SAR data are highly sensitive to the crop structure and dielectric properties [13], making them ideal for monitoring growth stages and crop development. By analyzing backscatter variations, SAR time-series data can track the entire rice growth cycle [14], distinguishing flooded fields (low backscatter during transplanting) from dry fields (higher backscatter) [15]. SAR-based paddy rice mapping has been extensively studied using sensors such as ENVISAT /ASAR [16–21], ALOS/PALSAR [22], RADARSAT/RADARSAT-2 [13,23,24], and TerraSAR-X [25–27]. The launch of Sentinel-1, a C-band SAR satellite constellation, has further enhanced the capability for large-scale paddy rice monitoring, providing high-temporal-resolution data every six days (Sentinel-1A and 1B) [28], making large-area paddy rice plant area mapping feasible [5,29]. Consequently, long-term Sentinel-1 SAR time-series data have been widely adopted for paddy rice mapping [5,30–32] and demonstrated as a stable SAR data source.

Accurate crop mapping, monitoring, and management require the extraction of spatial–temporal phenological information [6,33]. SAR-based paddy rice mapping typically utilizes backscatter characteristics, which capture both spatial and temporal features as key indicators [4,31,32,34]. Spatial features include texture, shape, and contextual information [35], while temporal features reflect backscatter variations associated with growth-stage-specific physiological changes [36]. The distinct backscatter variation patterns of rice fields differentiate them from other land cover types [35]. For high-temporal-resolution SAR data, features of paddy rice backscatter variation curves—such as maximum, minimum, mean, variance, curvature, and intervals between inflection points—can be extracted and used to construct phenological indicators [15]. These indicators facilitate paddy rice identification through thresholding, curve similarity methods [15,37], or the derived robust SAR-based Paddy Rice Index (SPRI) [38].

Traditional machine learning methods such as K-Nearest Neighbor (KNN), Decision Tree (DT), Support Vector Machine (SVM), and Random Forest (RF) have been applied in paddy rice mapping [12]. However, these approaches often struggle to fully exploit spatial–temporal information. Since the introduction of DL by Hinton et al. [39] in 2006, DL-based models have demonstrated superior performance in remote sensing crop identification.

Convolutional Neural Networks (CNNs), particularly Fully Convolutional Neural Network (FCN), U-Net [40], DeepLabv3+, PSPNet, and SegNet, have been widely employed for paddy rice segmentation, achieving higher classification accuracy [36,41]. However, CNN-based models primarily capture spatial patterns and are not optimized for time-series data [42]. To address this limitation, models such as 1D-CNN, Recurrent Neural Network (RNN) [43,44], Long Short-Term Memory (LSTM) [45–47], and BiLSTM [48,49] have been explored for time-series SAR-based mapping, with BiLSTM achieving the highest accuracy [12]. However, these methods do not fully exploit spatial information. Recently, 3D-CNN, ConvLSTM, and Transformer-based models have been developed to leverage both spatial and temporal variations for improved remote sensing image semantic segmentation [50,51], especially paddy rice mapping [52–54].

In recent years, some high-resolution paddy rice distribution maps have also been constructed, leveraging optical and SAR data for different regions, including Indonesia [46], Thailand [55], Nigeria [56], China [3,57,58], especially Northeast China [59], South China [60], and the Asian monsoon region [61]. Several studies have already focused on paddy rice mapping in Hainan Island. These studies utilized machine learning and DL methods to identify and delineate paddy rice fields [51,62,63]. However, these products can be further improved to fulfill the monitoring requirements for small and irregular ploughlands due to their limited spatial coverage and low spatial resolution.

Here, we proposed a paddy rice extraction and mapping framework using long-time-series SAR data for Hainan Island. BiLSTM- and LSTM-UNet-based models (named RiceLSTM and RiceTS) were proposed. The RiceLSTM model only uses the temporal information of SAR data, while the RiceTS model uses both the temporal and spatial information of SAR data. The remainder of this paper is organized as follows. Section 2 describes the study area and the dataset used in this study. Section 3 outlines the data processing workflow, as well as the proposed RiceLSTM and RiceTS models with their implementation. Section 4 presents the results. Section 5 discusses the influencing factors. Section 6 concludes the paper.

2. Study Area and Materials

2.1. Study Area

Hainan Island, located in the southernmost part of China (18°8′36.79″~ 20°9′48.71″N, 108°35′27.90″~111°4′16.78″E), covers an area of approximately 33,920 km^2 (Figure 1). The island's terrain exhibits a distinct elevational gradient, with higher elevations concentrated in the central region and progressively decreasing towards the periphery. The Five-Finger Mountain (1867 m) and Parrot Mountain (1811 m), both located near the island's center, represent the highest peaks. Moving outward from the central highlands, the landscape transitions through a series of descending mountains, hills, terraces, and plains, forming a concentric stratified landform (Figure 1c). Due to this topographic pattern, paddy rice cultivation is predominantly concentrated in the low-elevation peripheral regions of the island. Hainan Island encompasses diverse land cover types, including agricultural fields, urban areas, wetlands, grasslands, water bodies, and forests. The island's unique geographical and ecological characteristics provide an optimal environment for agricultural production, particularly for paddy rice cultivation.

Hainan Island experiences a tropical monsoon climate, characterized by high temperatures, abundant sunshine, and ample rainfall throughout the year. These climatic conditions create a favorable environment for rice cultivation, supporting multiple cropping cycles. Additionally, the island's fertile and diverse soil types supply essential nutrients for rice growth, further enhancing its agricultural productivity. Due to these advantages, Hainan Island has become one of China's key rice-producing regions, capable of sustaining two

or even three rice-growing seasons annually. The early paddy rice season extends from March to early July, while the late paddy rice season lasts from late July to late November. There is a slight difference in planting areas between the two seasons, with the late rice area typically exceeding that of the early crop [64].

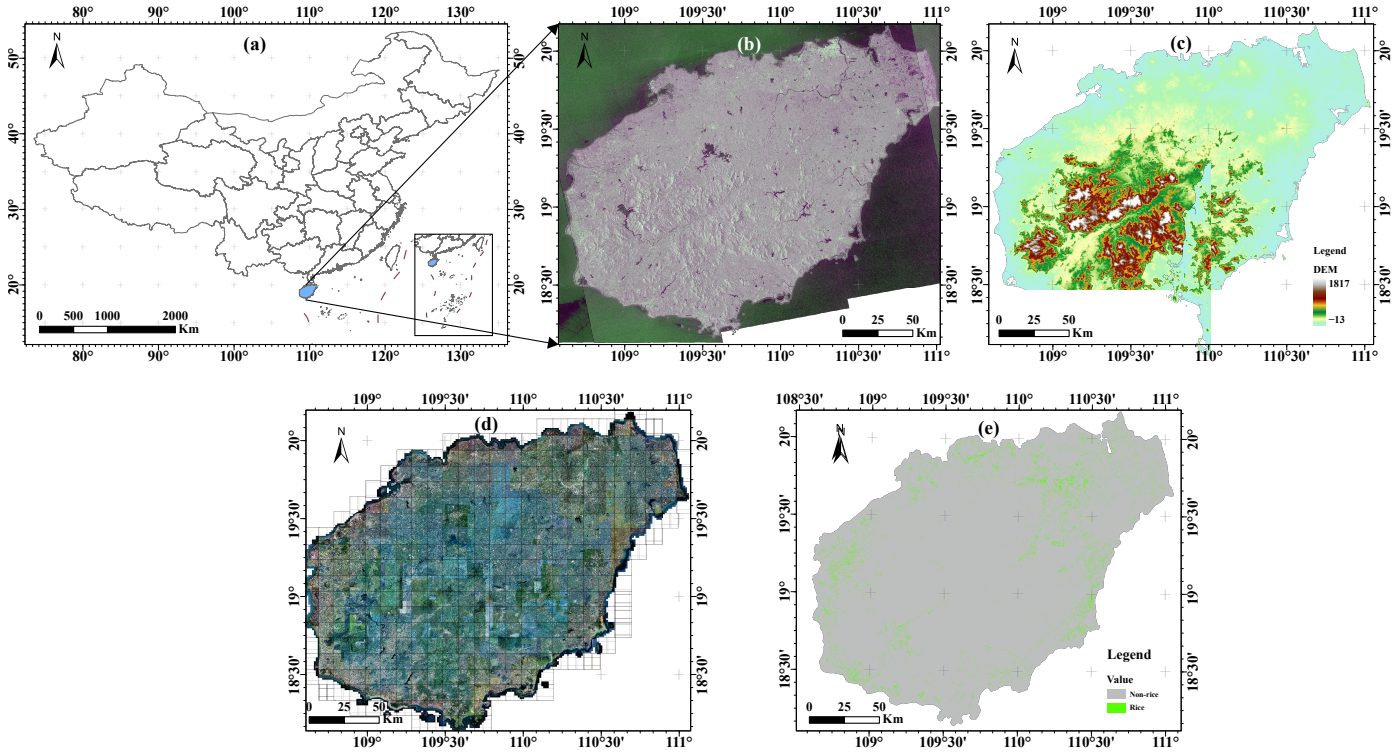


Figure 1. Spatial location and extent of the study area of Hainan Island. (**a**) Indicates the administrative divisions of China, highlighting the study area of Hainan Island with a blue polygon. (**b**) Indicates the Sentinel-1 SAR images (R: $\sigma_0(VH)$, G: $\sigma_0(VV)$, B: $\sigma_0(VH)$)from relative orbits of 55 (left bottom part), 157 (middle part), and 84 (top right part) acquired on 11, 6, and 13 January 2020. (**c**) Indicates the Copernicus Digital Elevation Model (DEM) data. (**d**) Indicates the Google Earth high-resolution images (true color composite) used to generate training samples. (**e**) Indicates the 10 m resolution paddy rice planting distribution product in 2020 from the Geographic Remote Sensing Ecological Network platform (http://www.gisrs.cn/shopdata?id=fa14d54e-19a6-4f35-b045-9dc166918060 (accessed on 19 July 2024)), which is also used to generate training samples.

2.2. Data

2.2.1. SAR Data

The frequent rainy and cloudy weather in Hainan Island limits the availability of high-quality cloud-free optical images from non-commercial platforms such as Landsat or Sentinel-2, making it challenging to obtain sufficient data for paddy rice mapping. To address this, 150 scenes of Level-1 Ground-Range-Detected (GRD) Sentinel-1A C-band (5.405 GHz) SAR data from the Interferometric Wide (IW) swath mode and VV/VH polarization were collected in 2020, along with 134 scenes in 2023, from Copernicus Open Access Hub (https://dataspace.copernicus.eu/, accessed on 18 April 2024). After Sentinel-1 slice assembly, a total of 90 effective observations were obtained for 2020 due to the assembly of two consecutive slices, resulting in 30 observations per orbit for relative orbits 157, 55, and 84 (Table A1). For 2023, 84 effective observations were obtained, with 30, 25, and 29 observations corresponding to relative orbits 157, 55, and 84, respectively (Table A1). All 174 observations were acquired in ascending mode. From Figure 2, we can see that the SAR data are primarily concentrated in the rainy season, aligning with the paddy rice planting period. Hainan Island is predominantly covered by Sentinel-1 SAR data from

relative orbit 157 (~90%), while the eastern and western regions are covered by relative orbits 55 and 84, respectively (Figure 1). In 2020, data from all three relative orbits were relatively complete, with only one missing acquisition from orbit 84 (Figure 2). However, in 2023, data gaps were more prominent, with five missing acquisitions from orbit 55, one from orbit 84, and one from orbit 157 (Figure 2). These data gaps pose certain challenges for time-series SAR-based paddy rice mapping. The next section will describe the methods used to address these gaps.

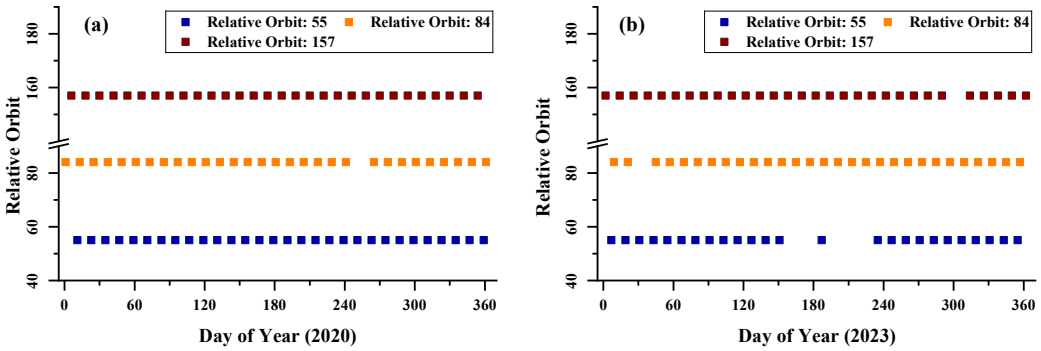


Figure 2. The temporal distributions (day of year) of Sentinel-1 SAR observations in 2020 (**a**) and 2023 (**b**).

2.2.2. Auxiliary Data

In this study, high-resolution Google Earth satellite images (Level 17, ~2 m) were also collected (Figure 1d) and used to identify the land cover types and the paddy rice fields for training. Meanwhile, the 10-meter-resolution paddy rice planting distribution map for Hainan Province in 2020, obtained from the Geographic Remote Sensing Ecological Network platform, was collected and utilized for training sample generation. The paddy rice mapping process followed a pixel classification strategy based on the planting and growth patterns of paddy rice in Hainan Province, achieving an overall accuracy of 85% (±2%) in 2020 (Figure 1e). Furthermore, a 10-meter-resolution land cover dataset (AIEC, https://engine-aiearth.aliyun.com/#/dataset/DAMO_AIE_CHINA_LC (accessed on 28 May 2024)) developed by DAMO AI Earth team was collected and used for training sample generation. This dataset integrates Sentinel-2 optical imagery and DL methods for detailed land cover classification.

3. Methodology

The flowchart of this study is presented in Figure 3. The workflow contains three main steps: SAR image preprocessing, paddy rice extraction using RiceLSTM and RiceTS models, and accuracy assessment. In the first step, long-time-series Sentinel-1 SAR σ^0 images are generated using the SARProcMod module. In the second step, two paddy rice mapping models, RiceLSTM and RiceTS, are proposed. To assess the performance of these models, artificial samples are selected for the confusion matrix and F1 score.

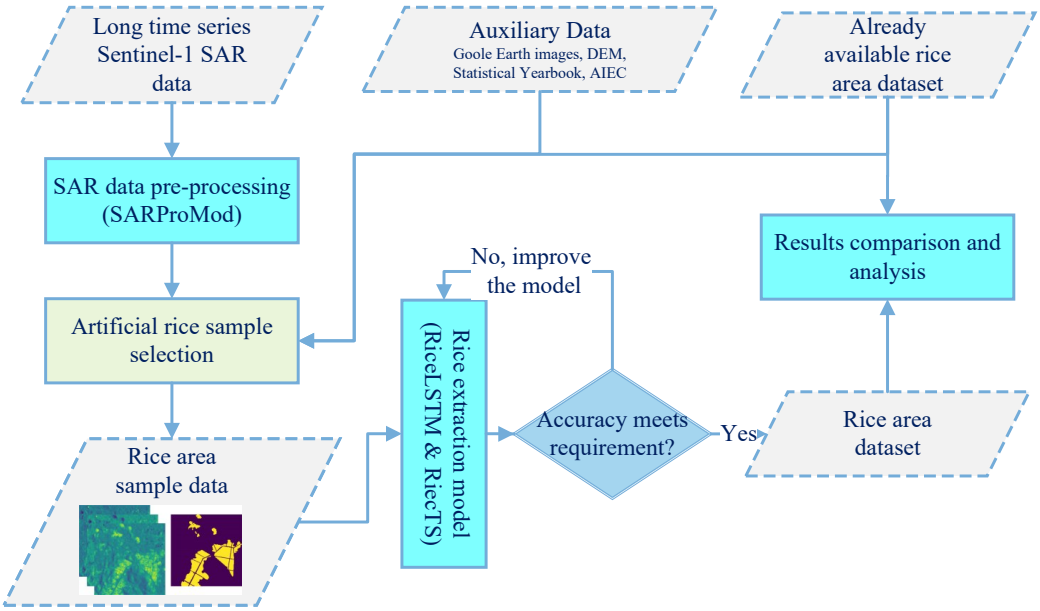


Figure 3. The flowchart showing the overall methods used in this study.

3.1. Data Preprocessing

Although VH polarization is more sensitive to rice growth stages, particularly during the flooding period [47], both VV and VH polarization data were utilized in this study to capture additional backscatter variations across non-rice land covers. The SARProcMod module proposed in [28] was used to preprocess Sentinel-1 SAR dataset (Figure 1c). The workflow included nine steps:

1. Thermal noise removal was applied to all Sentinel-1 SAR data.
2. Slice assembly was performed on consecutive slices when possible.
3. Orbit files were applied to update the orbit state vectors in the abstract metadata.
4. Radiometric calibration was conducted to generate radar backscatter bands (σ^0).
5. A Refined Lee filter was used to suppress speckle noise.
6. Multi-looking was applied to ensure square pixels.
7. Orthorectification of the radar backscatter bands was performed using the Range Doppler Terrain Correction algorithm with SRTM DEM data (spatial resolution: 3″)
8. The backscatter coefficient (in dB) was computed from the orthorectified radar backscatter band using the equation $10 \times \log_{10}(\sigma^0)$.
9. All Sentinel-1 SAR data were clipped and reprojected to a uniform UTM49N coordinate system, ensuring alignment with the study area extent.

Finally, the 10 m resolution Sentinel-1 SAR dataset was generated for paddy rice mapping. Additionally, Google Earth imagery, paddy rice planting distribution products, and land cover data underwent cropping, reprojection, and resampling to ensure spatial alignment with the Sentinel-1 SAR dataset.

3.2. RiceLSTM Model

The complex rice planting patterns and the heterogeneous paddy field environment pose significant challenges to accurately mapping paddy rice distribution using traditional methods. LSTM, known as a type of RNN, can effectively capture the temporal dependencies in time-series data [65]. BiLSTM, an extension of LSTM, processes sequences in both forward and backward directions, allowing it to capture contextual information more effectively [66]. Unlike Transformer, which relies on self-attention mechanisms and requires large datasets and high computational resources for effective training, LSTM and

BiLSTM offer a more efficient and robust solution for capturing temporal dependencies in time-series SAR data [67], making them suitable for paddy rice mapping. The proposed RiceLSTM model, illustrated in Figure 4, consists of two BiLSTM layers, each containing forward and backward layers (with two layers per direction) and a hidden size of 64, followed by an attention layer and a fully connected layer. The attention layer was introduced to make the model focus on key features and critical time steps, thereby enhancing its accuracy, interpretability, and robustness. This RiceLSTM model processes an input tensor with dimensions $256 \times 256 \times 15 \times 2$, where 256×256 represents the spatial dimensions of the image patches, 15 represents the number of observations, and 2 corresponds to the backscatter coefficient values from VV and VH polarization. The tensor was first reshaped into $65,536 \times 15 \times 2$ and fed into the BiLSTM layers, producing an output of size 128×15. The output is processed through an attention layer with 128 (64×2) neurons, followed by a final dense layer, which determines the paddy rice probability for each pixel. Finally, the output is reshaped back to 256×256 to match the input image dimensions. By analyzing the temporal variation of the backscatter coefficient of paddy rice and other land cover types, the RiceLSTM model effectively extracts key temporal features that differentiate paddy rice from non-rice areas.

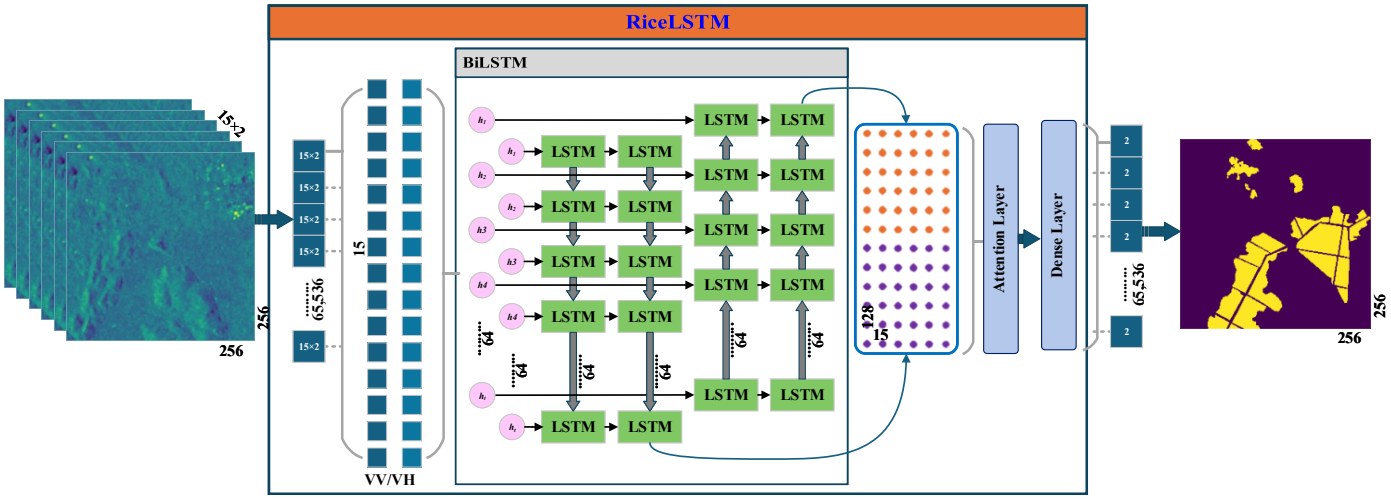


Figure 4. The flowchart of the RiceLSTM model used in this study. RiceLSTM is based on the BiLSTM model, with the number of features in the hidden state of 64 and the number of LSTM layers of 2. When the attention layer is applied, the entire BiLSTM output (15×128) is passed to the attention layer, which reduces it to a 1×128 tensor. The attention-enhanced output is then fed into the fully connected layer to estimate the paddy rice probability for each pixel. In the absence of the attention layer, only the last BiLSTM output sequence (1×128) is directly passed to the fully connected layer for classification, which produces a final output of 256×256, corresponding to the paddy rice classification map.

3.3. RiceTS Model

Although the proposed RiceLSTM model effectively captures the temporal dependencies in time-series data, it has inherent limitations. Specifically, RiceLSTM operates at the pixel level, thereby neglecting the spatial features present in remote sensing imagery. To address this limitation and fully exploit the spatial features of time-series SAR data, we introduce RiceMU (black dashed box in Figure 5), a Deep Convolutional Neural Network (DCNN) model designed for spatial feature extraction. RiceMU adopts the U-Net architecture [68], which features an encoder–decoder structure with skip connections to preserve fine-grained spatial details. The backbone of the U-Net model in RiceMU is MobileNetV2 [69], a lightweight CNN designed for computational efficiency. MobileNetV2 leverages depthwise separable convolutions to significantly reduce parameters and com-

putational costs while maintaining strong feature extraction capabilities. Additionally, it introduces inverted residuals and linear bottlenecks, which enhance feature expressiveness and prevent information loss during non-linear transformations. The combination of U-Net and MobileNetV2 enables RiceMU to efficiently process SAR image patches and extract fine-grained spatial features for paddy rice mapping. RiceMU processes an input tensor of dimensions $256 \times 256 \times d$, where 256×256 represents the spatial dimensions of the image patches, and d (in this study of 30 (15×2)) denotes the number of feature channels. The model undergoes five levels of downsampling through convolutional layers to capture hierarchical spatial features, followed by upsampling operations to restore spatial resolution while retaining essential details. The final output is a $256 \times 256 \times 2$ probability map, where each pixel is assigned a likelihood of being classified as paddy rice.

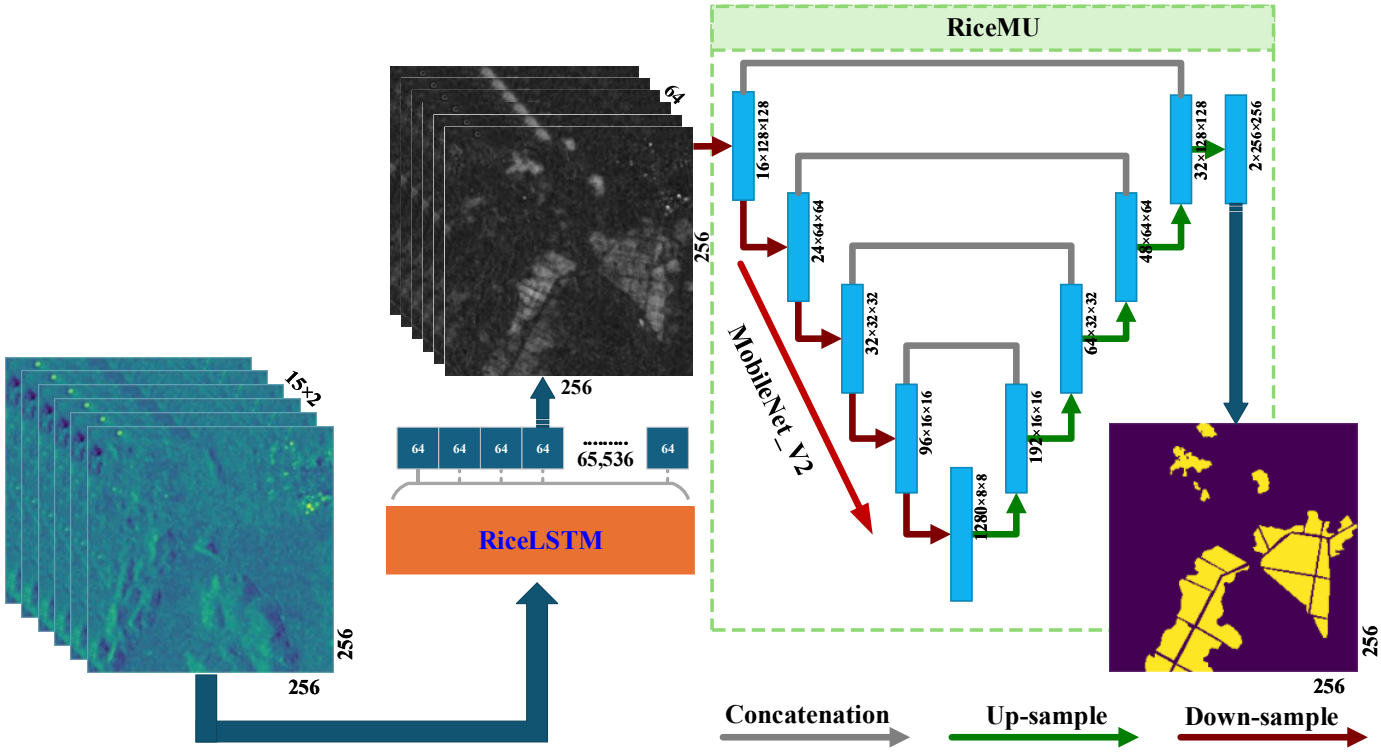


Figure 5. The flowchart of the RiceTS model used in this study. The RiceLSTM layer is based on RiceLSTM model (Figure 4) and generates an output size of 65,536 ×64. This output is reshaped into a $256 \times 256 \times 64$ feature map and subsequently fed into the RiceMU layer (green dashed box), which processes the spatial features and produces a final output of 256×256, corresponding to the paddy rice classification map.

To fully exploit and utilize the temporal and spatial features embedded in time-series SAR data, many DL models integrating LSTM and CNN have been proposed, employing different sequences for extracting spatial and temporal features. Here, we proposed a RiceTS model, which first extracts temporal features using RiceLSTM layer and then spatial features afterward using RiceMU layer. The proposed RiceTS model, illustrated in Figure 5, consists of two main components:

1. RiceLSTM layer (temporal feature extraction): The first stage of RiceTS employs RiceLSTM model as a layer to capture temporal dependencies in time-series SAR data. The workflow of the RiceLSTM layer follows that of the RiceLSTM model but generates a feature representation of size 65,536 ×64.
2. RiceMU layer (spatial feature extraction): The temporal feature output from the RiceLSTM layer is reshaped into a spatial format of $256 \times 256 \times 64$ and passed into the

RiceMU layer. This step allows the spatial encoder–decoder network to further refine the feature representation and perform detailed segmentation of paddy rice regions.

The final output of RiceTS model has a shape of 256 × 256 × 2, which represents the probability distribution of paddy rice presence at each pixel. By separating spatial and temporal feature extraction into distinct components, RiceTS allows for a comprehensive analysis of their relative importance in paddy rice mapping. Furthermore, RiceMU can be independently applied to assess the effectiveness of spatial feature extraction alone, enabling a comparative evaluation of spatial and temporal feature contributions in SAR-based rice area extraction.

3.4. Training Sample Generation

The training samples for the RiceLSTM and RiceTS models were labeled at the pixel level, with attribute values assigned as 1 for paddy rice and 0 for non-rice. Firstly, all Sentinel-1 SAR images from 2020 with the relative orbit of 155, along with Google Earth image (Figure 1d), and the paddy rice distribution map (Figure 1e) were stacked together to aid in generating training and validation samples. Through visual analysis of high-resolution optical images and the paddy rice distribution map, 490 paddy rice field polygons were delineated within the study area. Considering the 10 m resolution of the imagery, patches were cropped to a size of 256 × 256 using a sliding window to better capture spatial features of paddy rice and non-rice for model training. However, the total paddy rice planting area in Hainan Island is approximately 1.10×10^5 ha, accounting for only 3.3% of the island's total area (3.392×10^6 ha). Moreover, not all paddy rice fields were included during sample generation, further exacerbating the issue of severe class imbalance. To address this, patches with fewer than 100 paddy rice pixels were discarded, resulting in a final set of 896 sample patches. For the RiceLSTM model, 973,124 pixels for paddy rice and 2,000,000 non-rice pixels were randomly selected from the 896 patches, yielding a total of 2,973,124 samples. These samples were split into 80% for training and 20% for validation. For the RiceTS model, all 896 patches were used, maintaining the same 80%–20% split for training and validation. To enhance sample diversity and variability, data augmentation techniques such as random vertical and horizontal flip were applied during training. To mitigate class imbalance, a higher weight was assigned to the paddy rice class in the loss function, penalizing the model more for misclassifications of minority samples. The weights for the non-rice and paddy rice were defined as $\left[\sqrt{C_{Rice}}/(\sqrt{C_{Non}}+\sqrt{C_{Rice}}), \sqrt{C_{Non}}/(\sqrt{C_{Non}}+\sqrt{C_{Rice}})\right]$, where C_{Rice} and C_{Non} represent the number of rice and non-rice pixels in the patches, respectively.

3.5. Validation Metrics

The evaluation of a paddy rice extraction model involves the application of various statistical metrics. In this study, Accuracy, Precision, Recall, F1 score, and Cohen's Kappa coefficient (k) were utilized to assess model performance (Equations (1)–(6)). Here, TP, FP, TN, and FN represent true positive, false positive, true negative, and false negative, respectively.

$$Accuracy = \frac{TN + TP}{TN + FP + FN + TP} \tag{1}$$

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{4}$$

$$k = \frac{Accuracy - P_e}{1 - P_e} \tag{5}$$

$$\text{where } P_e = \frac{(TP + FN)(TP + FP) + (TN + FN)(TN + FP)}{(TP + TN + FP + FN)^2} \tag{6}$$

3.6. Experimental Settings

These two models were implemented using Python 3.10, Pytorch 2.2, and CUDA 12.1. Training and evaluation were conducted on a workstation equipped with NVIDIA RTX 8000 GPU (48 GB memory), two Intel(R) Xeon(R) Gold 6134 CPUs, 512 GB memory, a 512 GB SSD, and an 8 TB HDD. Through trial and error in the training phase, batch sizes of 65,536, 12, and 4 were determined for RiceLSTM, RiceMU, and RiceTS models, respectively. The models were trained for 200 epochs with an early stopping mechanism set at a patience of 20 epochs to prevent overfitting. We employed the *CrossEntropy* loss function, the *Adam* optimizer with a constant learning rate of 0.001, and used *accuracy* as the evaluation function. The most optimal model was ultimately preserved. For post-processing, a threshold value of 0.5 was applied to convert probability values into binary classifications.

4. Results

4.1. The Backscatter Variations for Different Land Cover Types

After preprocessing all the Sentinel-1 SAR data, the SAR images from 2020 were utilized to analyze the temporal variations in backscatter for paddy rice and non-rice crop types. Five land cover types—paddy rice, buildings, water, dryland, and forest—were selected for time-series curve analysis, with a total of 20 polygons chosen as representative samples. The VV and VH backscatter coefficients for each land cover type were averaged at each time step, yielding 30 pairs of values, as shown in Figures 6 and 7. These figures indicate that the temporal backscatter distribution patterns differ significantly among the land cover types. Regardless of the land cover type, the backscatter in VH polarization is consistently lower than that in VV polarization (Figures 6 and 7). Water bodies exhibit lower backscatter due to specular reflection of radar signals (Figure 6), while buildings demonstrate stronger backscatter as a result of double scattering, and forests exhibit high backscatter due to volume scattering (Figure 6). These three land cover types display relatively stable backscatter over time, indicating their temporal consistency (Figure 7). In contrast, paddy rice fields and drylands exhibit lower backscatter with greater temporal variation, indicating their dynamic seasonal changes (Figure 7).

In Figure 6, the backscatter signatures of paddy rice and dryland vegetation exhibit a significant degree of overlap. However, Figure 7 clearly reveals distinct differences in their temporal variations. For paddy rice, two distinct planting cycles are observed, separated by early July, whereas dryland vegetation does not exhibit such cyclic behavior. As paddy rice grows, backscatter increases, reaching its peak between late May and early June. During the maturity stage, backscatter gradually decreases, followed by a sharp decline during the harvesting period in early July. Subsequently, the growth cycle of late paddy rice begins with an agricultural flooding period in mid-July, during which the backscatter of paddy rice closely resembles that of water. Figures 6 and 7 demonstrate that time-series Sentinel-1 SAR data are effective for paddy rice mapping. To better capture the distinct phenological patterns, the 30 observation datasets were divided into two groups, each containing 15 observations, corresponding to the growth cycles of early and late paddy rice.

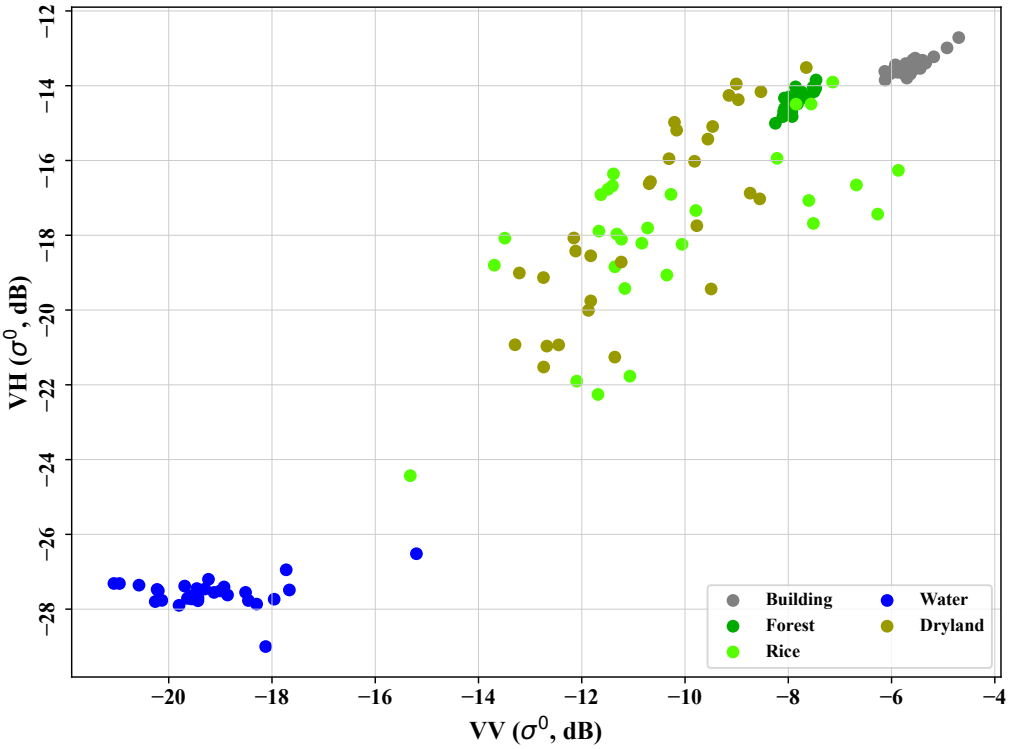


Figure 6. The time-series VV and VH polarization backscatter distribution map for different land cover types.

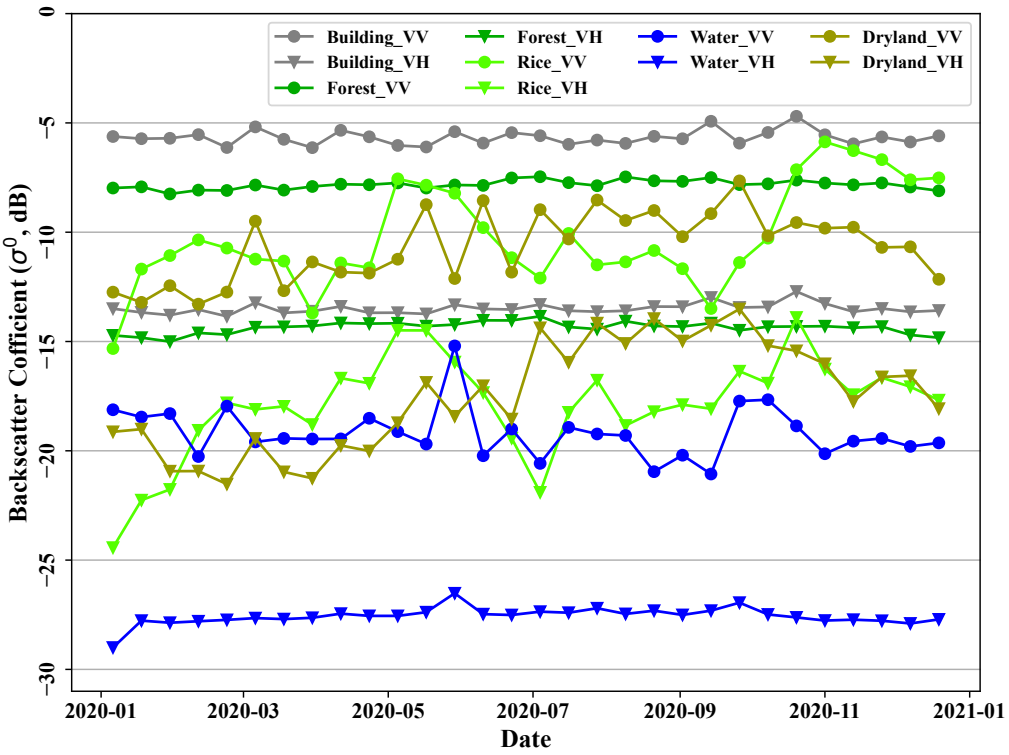


Figure 7. The variations in backscatter from Sentinel-1 SAR data for different land cover types and VV and VH polarization.

4.2. Comparison Between RiceLSTM and RiceTS

To evaluate and compare the RiceLSTM (with and without an attention layer) and RiceTS models, the training and validation samples in 2020 were used. The five evaluation metrics outlined in Section 3.5 were employed for quantitative assessment. The RiceLSTM-Attention model was configured with the same BiLSTM layers and parameter settings as the RiceLSTM (without attention) model. The RiceMU model was also configured to evaluate the significance of spatial features. By utilizing stacked training samples as an input (Table A2), it inherently incorporates temporal variations in SAR imagery. This enables RiceMU to exploit both spatial and temporal characteristics, offering a complementary perspective for assessing paddy rice classification performance. For the RiceTS model, the temporal feature extraction layer (BiLSTM layer) was configured identically to that of the RiceLSTM-Attention model, while the spatiotemporal feature extraction layer (RiceMU layer) was designed following the architecture of the RiceMU model.

The quantitative results are presented in Table 1 and Figure 8. The RiceLSTM-Attention model achieved accuracies of 0.9182 (+2.1%) and 0.9245 (+1.8%) for early and late paddy rice, respectively, demonstrating a notable improvement over the RiceLSTM model (0.8971 and 0.9067). The results indicate that the attention mechanism significantly enhances the performance of RiceLSTM across all the metrics, particularly in capturing subtle variations in paddy rice field dynamics. The RiceMU model demonstrated competitive performance, achieving high classification accuracy (0.9642 for early rice and 0.9604 for late rice), with recall values exceeding 0.9500. However, its precision was slightly lower than that of RiceTS, particularly for late rice (0.7474), indicating a tendency to classify non-rice areas as rice. The RiceTS model further improves in classification accuracy, achieving 0.9656 (+4.74%) and 0.9808 (+5.63%) for early and late paddy rice, respectively, surpassing the RiceLSTM-Attention and RiceMU models. However, the RiceTS model exhibits lower precision (0.7512 for early and 0.8465 for late paddy rice) compared to the RiceLSTM-Attention model, indicating a higher rate of false positives. This overestimation may result from similarities in backscatter patterns between paddy rice fields and non-rice land cover types, leading to the misclassification of non-rice regions as paddy rice fields.

Table 1. Accuracy assessment for classification results generated from different models for early and late paddy rice in 2020.

Model	Accuracy	Precision	Recall	F1 Score	Kappa
RiceLSTM (Early)	0.8971	0.8636	0.8142	0.8382	0.7628
RiceLSTM (Late)	0.9067	0.8650	0.8472	0.8560	0.7870
RiceLSTM With Attention (Early)	0.9182	0.8839	0.8635	0.8735	0.8131
RiceLSTM With Attention (Late)	0.9245	0.8809	0.8898	0.8853	0.8291
RiceMU (Early)	0.9642	0.7787	0.9517	0.8566	0.8364
RiceMU (Late)	0.9604	0.7474	0.9812	0.8484	0.8264
RiceTS (Early)	0.9656	0.7512	0.9285	0.8305	0.8116
RiceTS (Late)	0.9808	0.8465	0.9812	0.9088	0.8982

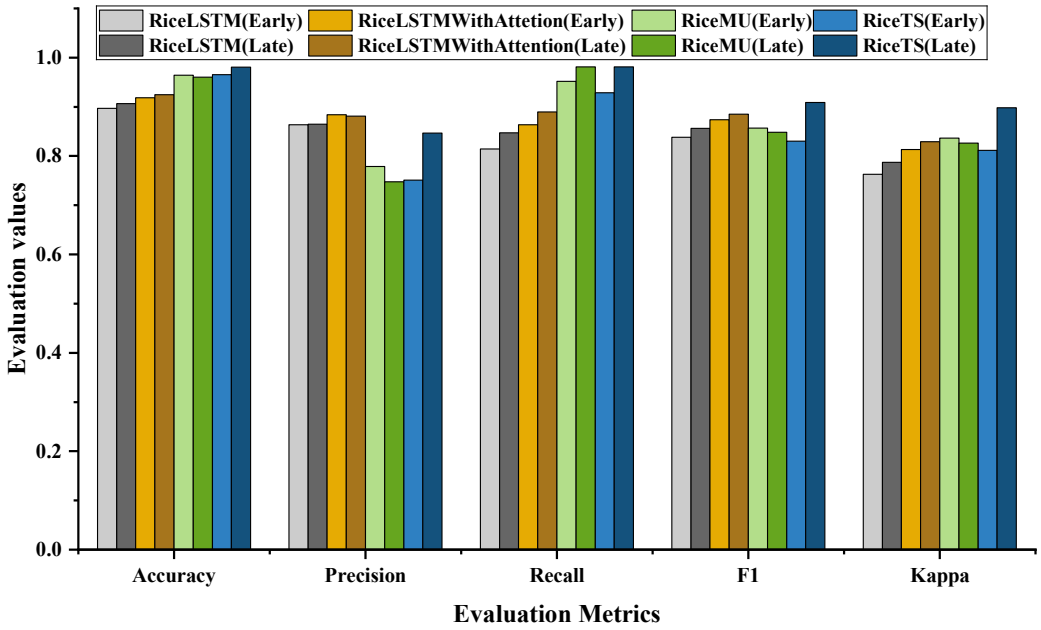


Figure 8. The evaluation indices for the classification results of the proposed models for early and late paddy rice in 2020.

Figure 9 presents the classification results of the proposed models for the test area. For test area (1), the RiceLSTM-Attention, RiceMU, and RiceTS models achieved favorable classification performance, effectively capturing the distribution of paddy rice fields. However, a significant omission of paddy rice is observed in the B1 region (blue box B1) in the paddy rice product (Figure 9b), whereas the RiceLSTM, RiceMU, and RiceTS models successfully identify paddy rice in this area. Both the paddy rice distribution product (Figure 9b) and the RiceLSTM model result (Figure 9c) exhibit salt-and-pepper noise due to pixel-based classifications. The paddy rice distribution product, generated using Sentinel-2 optical imagery, contains salt-and-pepper noise, while the RiceLSTM-Attention model result, derived from SAR data, exhibits even more pronounced noise (blue box B2). For test area (2), the RiceLSTM-Attention model struggles to effectively extract the spatial distribution of paddy rice, whereas the RiceMU and RiceTS models achieve more accurate extraction, highlighting the importance of spatial features in paddy rice mapping. Additionally, the RiceTS model demonstrates superior continuity in the extracted results compared to RiceMU, further highlighting the significance of temporal features. Across both test areas, the RiceTS model effectively suppresses speckle noise, improving overall classification accuracy. However, this suppression comes at the cost of losing some fine details as certain roads or ridges are misclassified as rice (blue box), leading to a reduction in precision. Nevertheless, the extracted paddy rice boundaries from the RiceLSTM-Attention and RiceTS models closely align with the actual remote sensing imagery.

These results indicate that the RiceTS model excels in capturing the temporal and spatial characteristics of rice growth, resulting in improved accuracy in paddy rice mapping. Therefore, the RiceTS model will be adopted for paddy rice mapping and analysis in the subsequent sections.

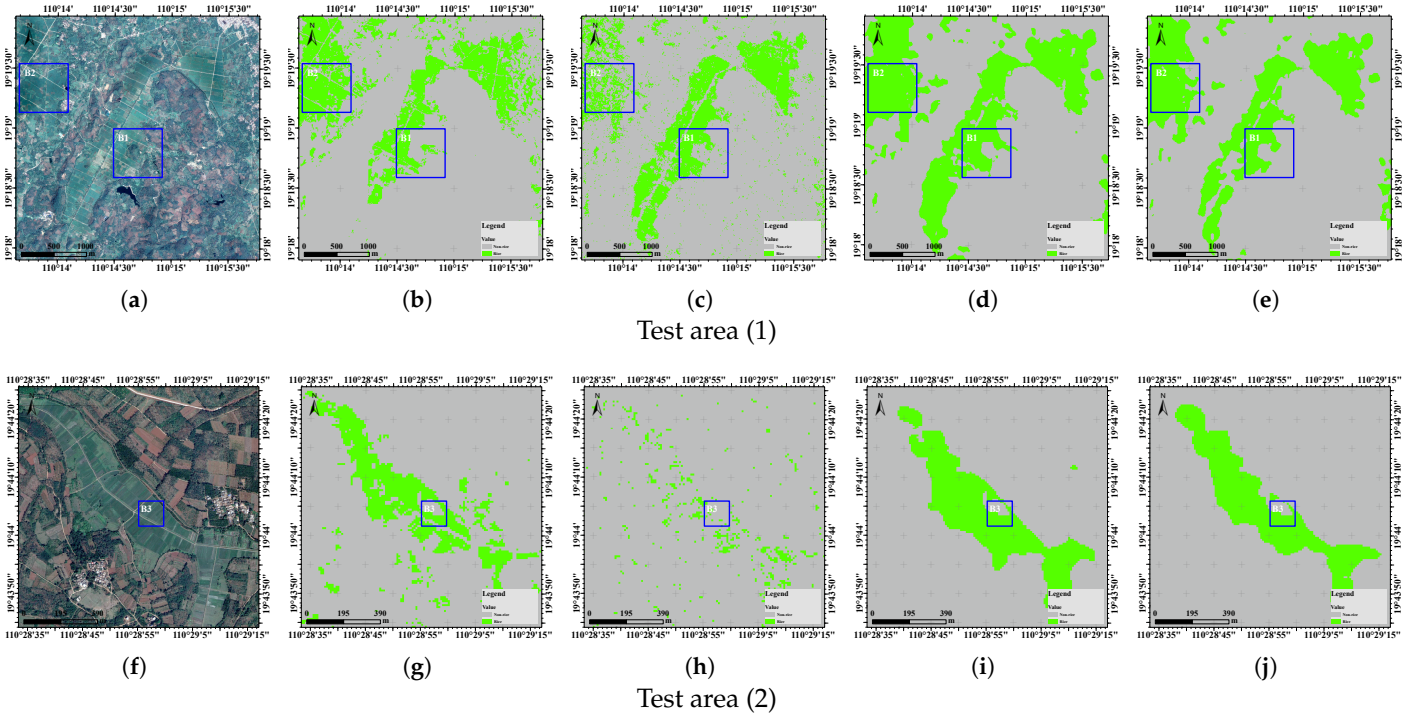


Figure 9. The comparison between the paddy rice product, the results derived from the RiceLSTM-Attention, RiceMU, and RiceTS models, (**a**,**f**) for GEE high-resolution image (true color composite), (**b**,**g**) for the paddy rice product (Figure 1e), (**c**,**h**) for the result derived from RiceLSTM-Attention model, (**d**,**i**) for the result derived from RiceMU model, and (**e**,**j**) for the result derived from RiceTS model.

4.3. Paddy Rice Distribution Map

Figure 10 illustrates the early and late paddy rice distribution map in 2020, as derived from the proposed RiceTS model. According to the model, the early and late paddy rice areas were 1.123×10^5 ha and 9.037×10^4 ha , respectively, resulting in a total of 2.027×10^5 ha in 2020. In comparison, statistical data from the *Hainan Statistic Yearbook* (2021), published by China Statistics Press, reported early and late paddy rice areas of 1.100×10^5 ha and 1.284×10^5 ha, respectively, with a total of 2.384×10^5 ha. The extracted annual paddy rice planting area achieved an accuracy of 90.6%, demonstrating good consistency with the official statistics.

The differences between the classified paddy rice area and the statistical data were 2.3×10^3 ha (2.1%) for early paddy rice and 3.80×10^4 ha (29.6%) for late paddy rice. The relatively lower extraction accuracy for late rice is primarily due to the training samples derived from early rice. Conversely, this also indicates that the model exhibits a certain degree of temporal transferability.

As shown in Figure 10, paddy rice fields are rarely distributed in the central mountainous region of Hainan Island due to significant terrain fluctuations. In contrast, large-scale rice cultivation is primarily concentrated in the northern and northeastern regions, as well as along the southern coastal areas. Additionally, double-cropping paddy rice is widely distributed in the eastern part of Hainan Island, accounting for approximately 32.5% of the total paddy rice area. Early-season single-crop rice is also prevalent in this region, comprising approximately 26.9% of the total paddy rice area. Meanwhile, in the western part of the island, large-scale single-crop late rice dominates, accounting for approximately 40.6% of the total paddy rice area.

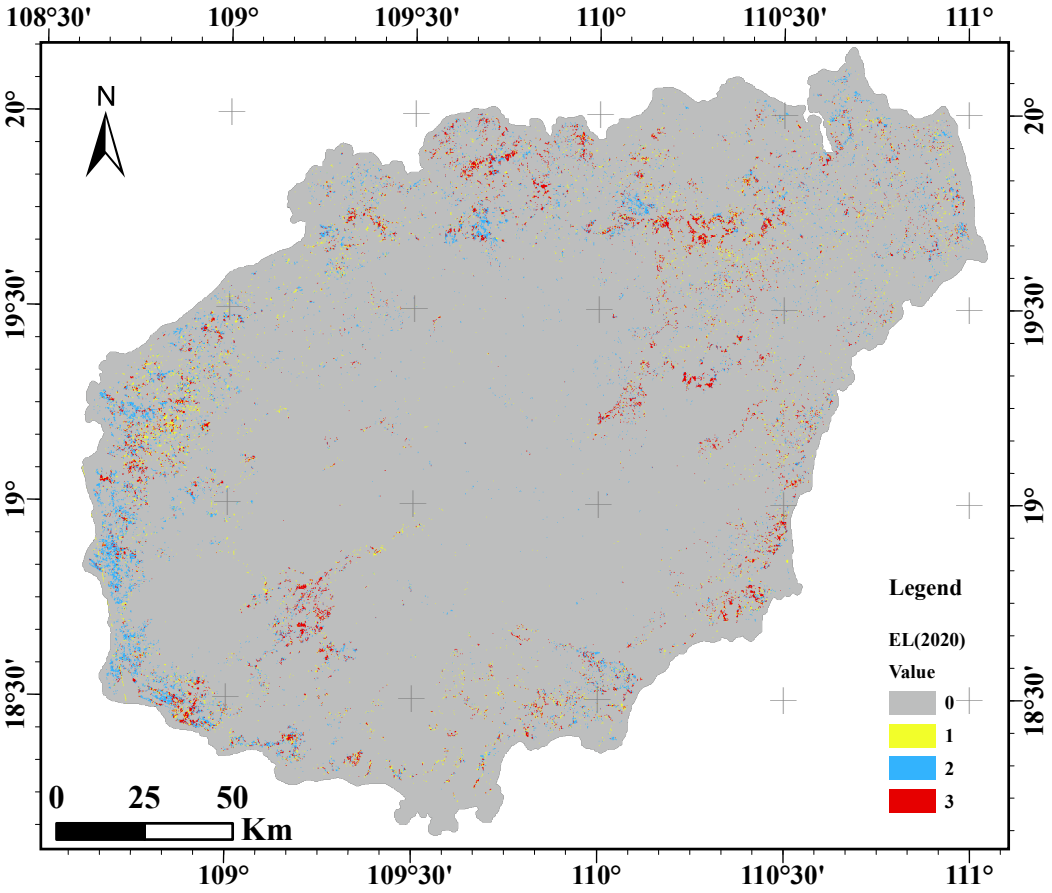


Figure 10. The paddy rice distribution map in 2020 derived from the RiceTS model: 0 for non-rice area, 1 for only late paddy rice, 2 for only early paddy rice, and 3 for both.

For paddy rice mapping in 2023, the RiceTS model was employed. However, this model requires continuous time-series data as the input, and the dataset for 2023 was incomplete (Figure 2, Table A1), making the direct application of the model unfeasible. To address the issue of missing data, a data imputation approach was implemented. For data from relative orbit 157, five missing acquisitions between 19 June and 6 August 2023 contained only slice 1. The gap was filled using SAR data from the corresponding period in 2020. Similarly, for data from relative orbit 55, the five missing acquisitions (12 June, 24 June, 18 July, 30 July, and 11 August) were also supplemented with SAR data from the same periods in 2020. For relative orbit 84, the missing acquisition on 12 February was reconstructed by averaging the data from 21 January and 14 February 2023. This imputation approach resulted in a complete time-series dataset (30 observations) covering the entire early and late paddy rice growth cycles for 2023.

Using the model trained in 2020, the paddy rice was mapped for 2023, as shown in Figure 11. Due to the unavailability of statistical data and paddy rice distribution products for validation, the classification accuracy could not be directly assessed. According to the classification results, the estimated early paddy rice area is 1.284×10^5 ha, while the late paddy rice area is 9.037×10^4 ha, showing no significant change compared to 2020. The spatial distribution patterns of early rice, late rice, and double-cropping paddy rice in 2023 are consistent with those observed in 2020, further demonstrating the robust spatiotemporal transferability of the proposed model.

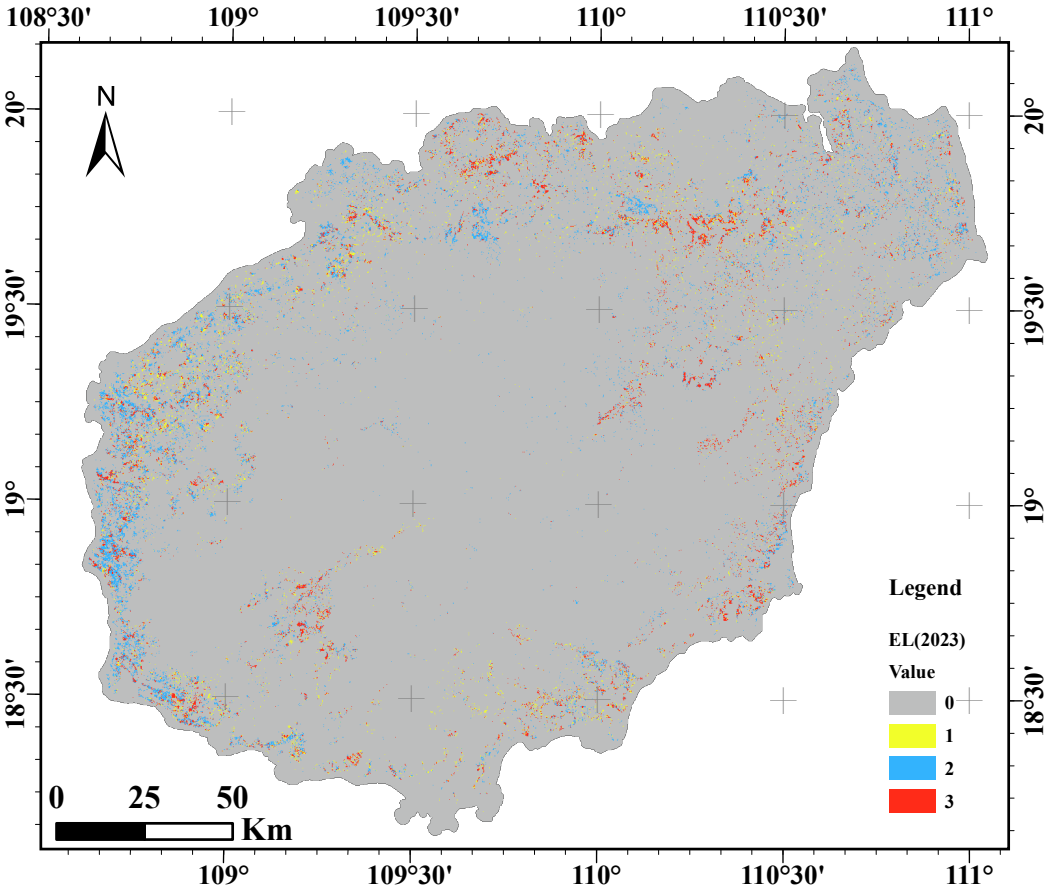


Figure 11. The paddy rice distribution map in 2023 derived from the RiceTS model: 0 for non-rice area, 1 for only late paddy rice, 2 for only early paddy rice, and 3 for both.

5. Discussion

This study aims to investigate the effective use of time-series SAR data for extracting paddy rice areas in Hainan Island. Previous studies have demonstrated the efficacy of integrating temporal, spatiotemporal, and spatiotemporal–spectral feature fusion for paddy rice mapping [12,45,52,53,65,66]. Two models were developed for paddy rice mapping: RiceLSTM and RiceTS. RiceLSTM utilizes a BiLSTM network with an attention mechanism, achieving classification accuracies of 0.9182 (early rice) and 0.9245 (late rice). RiceTS, which integrates a U-Net with MobileNetV2, further improves the classification accuracy to 0.9656 (early rice) and 0.9808 (late rice). The extraction accuracy in this study exceeds 90%, aligning with the performance levels reported in the existing research [5,12,48]. Compared to the 2020 statistics, the estimated areas differed by 2.1% for early rice and 29.6% for late rice, highlighting both the model's accuracy and robustness. Using the RiceTS model, the paddy rice areas in Hainan were successfully extracted from time-series Sentinel-1A SAR data for both 2020 and 2023. When applied to the 2023 SAR data, the classified paddy rice distribution exhibited spatial patterns that are consistent with 2020, demonstrating strong spatiotemporal transferability. These findings confirm that integrating time-series SAR imagery with DL effectively captures both spatial and temporal features, significantly enhancing paddy rice mapping accuracy. This underscores the model's potential for large-scale paddy rice monitoring in regions with similar agricultural practices. Hainan Island's warm and humid climate supports paddy rice cultivation, including single-, double-, and, in some areas, triple-season cropping. Double-season rice, which constitutes approximately 30% of the total paddy rice area, is predominant in the eastern region, whereas single-season late rice is primarily cultivated in the western region.

However, some challenges remain in paddy rice mapping using time-series SAR data. The completeness of time-series data is crucial for accurately mapping paddy rice cultivation areas, particularly for models that rely on temporal backscatter features. Missing data disrupt temporal continuity, reducing classification reliability and accuracy. Accurate paddy rice mapping depends on capturing key phenological stages, such as transplanting, tillering, and maturity [15]. Data gaps during these critical periods hinder the model's ability to fully capture paddy rice growth dynamics, often leading to misclassification—particularly in distinguishing paddy rice from other vegetation with similar backscatter characteristics, such as dryland crops or wetland vegetation [70]. Most studies have not directly addressed data gaps in time-series SAR data for paddy rice mapping [12,48]. Shen and Nie attempted to mitigate this issue by reconstructing the time-series SAR dataset using a grouping strategy, but this approach also introduced challenges in accurately capturing paddy rice backscatter variations [14]. In this study, data interpolation and reuse methods were applied to address missing data, although these techniques introduced a certain level of uncertainty in paddy rice mapping.

Compared to optical data, SAR data are more susceptible to salt-and-pepper noise [12], which can negatively impact the accuracy of paddy rice mapping. The precise identification of paddy rice fields depends on detecting the subtle variations in backscatter associated with different growth stages [12]. However, speckle noise can obscure these variations, leading to the misclassification of paddy rice fields or their misinterpretation as other land cover types with similar backscatter properties. Additionally, speckle noise complicates feature extraction, particularly for small or heterogeneous paddy rice fields. To mitigate these effects, a 3D-Gamma adaptive filter and the Savitzky and Golay (S–G) smoothing method were applied to process time-series SAR data in both the spatial and temporal domains, facilitating the identification of long-term change trends [12].

DL models require a sufficient number of high-quality training samples to achieve accurate and generalizable results. Sample generation plays a crucial role in the accuracy and reliability of paddy rice mapping using remote sensing data and DL models [63]. The relatively lower extraction accuracy for late rice underscores the importance of complete and representative training samples. While the existing paddy rice distribution products [62,63] covering Hainan Island can be used for training sample generation, their limited temporal and spatial coverage, along with inherent accuracy constraints, may introduce biases into the training data [63]. Field surveys and manual interpretation remain essential but are time-consuming and labor-intensive [48]. Furthermore, the relatively small proportion of paddy rice cultivation areas can lead to sample imbalance, causing the models to favor majority classes and reducing sensitivity to paddy rice fields. Addressing sample generation challenges is thus a critical aspect of improving DL-based paddy rice mapping.

Recent changes in the paddy rice transplantation methods in Hainan [64] have also influenced the backscatter characteristics observed in time-series Sentinel-1 SAR data, potentially impacting mapping accuracy. Additionally, variations in transplanting times across regions and seasons further complicate classification accuracy, particularly when using time-series SAR data. Another persistent challenge is the misclassification of wetland vegetation as paddy rice due to their similar backscatter characteristics, temporal backscatter variations, vegetation structures, and geographical overlap. The similarity in backscatter signals makes it difficult to accurately distinguish between paddy rice and wetland vegetation, increasing the likelihood of misclassification in paddy rice mapping [70].

The long-term SAR paddy rice mapping process requires substantial computational power for data preprocessing. In this study, we collected a total of 284 scenes of Sentinel-1 SAR data. Using the SARProcMod module, the download time for each scene of Sentinel-1

SAR (approximately 1 GB) is approximately 2 min. The SARProcMod module was also used for processing SAR data, with each scene requiring approximately 20 min (preprocessing and clip). The total preprocessing time amounted to around 95 h, resulting in a 10 m resolution dataset with a total disk volume of around 500 GB. Many studies have leveraged the Google Earth Engine (GEE) platform, which offers direct access to preprocessed Sentinel-1 SAR data [71], significantly reducing the time required for data preprocessing [8,9,62]. However, the GEE platform does not support end-to-end processing of SAR data using deep learning frameworks, and it imposes strict limitations on downloading large-scale Sentinel-1 datasets. DL training also demands significant GPU resources (Table A2), which impacts the model's scalability and practical deployment. The total parameters for the two proposed models reach up to 151,042 and 3,133,066 separately, requiring 24 GB and 48 GB of GPU memory during training, respectively. This high computational demand makes it challenging to deploy these models on edge devices or in resource-constrained environments, limiting their practical application in real-time paddy rice mapping. Recently, Transformer [67], particularly Swin Transformer (Shifted Window Transformer) [50], has garnered significant attention for its effectiveness in remote sensing image analysis. Swin Transformer employs a window-based self-attention mechanism, and its hierarchical structure enables it to capture multi-scale features. This capability is particularly beneficial for extracting both spatial and temporal features at different scales, making it highly suitable for rice field identification across various growth stages. Swin Transformer and its derivatives are increasingly being applied to paddy rice and other crop information extraction [51,72] due to their high efficiency and low system requirements.

6. Conclusions

In this study, we proposed two DL models, RiceLSTM and RiceTS, for accurate paddy rice field extraction from time-series SAR data in Hainan Island. Both models demonstrate the potential of leveraging temporal features in SAR data for effective paddy rice mapping. RiceLSTM, particularly with an attention mechanism, significantly outperforms the traditional LSTM model in terms of accuracy. RiceMU leverages spatiotemporal information by using stacked inputs, leading to improved extraction accuracy compared to RiceLSTM. RiceTS, on the other hand, highlights the effectiveness of temporal convolutional networks in capturing spatial dynamics, achieving a higher F1 score than RiceLSTM. The integration of multi-temporal SAR data effectively captures paddy rice growth dynamics, enhancing the detection of both early- and late-season paddy rice. This approach provides a reliable solution for paddy rice mapping in regions with frequent cloud cover and heavy rainfall.

However, the results also emphasize the dependence of these models on the completeness and accuracy of the training data. Issues such as sample mislabeling, incomplete representation of paddy rice field variability, and temporal data gaps may limit the models' generalization ability and overall accuracy. Despite these challenges, the models demonstrate strong potential for spatiotemporal generalization across different years, underscoring their adaptability for large-scale crop mapping applications.

Future research will focus on improving the sample generation strategies, optimizing the model efficiency, effectively addressing data gaps, mitigating sample imbalance, and preserving edge details in DL-based paddy rice mapping. These efforts will contribute to improving the robustness and applicability of DL models for agricultural monitoring using SAR data.

Author Contributions: G.S. proposed and designed the technique road map, contributed to the creation of the algorithmic methods, and collected and processed the SAR data. J.L. helped with the results analysis and reviewed the manuscript. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement: The Sentinel-1 SAR data listed in Table A1 are available in https://dataspace.copernicus.eu/ (accessed on 18 April 2024). The other data presented in this study are available on request from the corresponding author.

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Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A

Table A1. SAR data used in this study.

Year: 2020, Relative Orbit: 157								
No.	**Acquisition Date**	**Slice**	**No.**	**Acquisition Date**	**Slice**	**No.**	**Acquisition Date**	**Slice**
1	2020/1/6	1, 2	11	2020/5/5	1, 2	21	2020/9/2	1, 2
2	2020/1/18	1, 2	12	2020/5/17	1, 2	22	2020/9/14	1, 2
3	2020/1/30	1, 2	13	2020/5/29	1, 2	23	2020/9/26	1, 2
4	2020/2/11	1, 2	14	2020/6/10	1, 2	24	2020/10/8	1, 2
5	2020/2/23	1, 2	15	2020/6/22	1, 2	25	2020/10/20	1, 2
6	2020/3/6	1, 2	16	2020/7/4	1, 2	26	2020/11/1	1, 2
7	2020/3/18	1, 2	17	2020/7/16	1, 2	27	2020/11/13	1, 2
8	2020/3/30	1, 2	18	2020/7/28	1, 2	28	2020/11/25	1, 2
9	2020/4/11	1, 2	19	2020/8/9	1, 2	29	2020/12/7	1, 2
10	2020/4/23	1, 2	20	2020/8/21	1, 2	30	2020/12/19	1, 2
Year: 2020, Relative Orbit: 55								
No.	**Acquisition Date**	**Slice**	**No.**	**Acquisition Date**	**Slice**	**No.**	**Acquisition Date**	**Slice**
31	2020/1/11	5, 6	41	2020/5/10	5, 6	51	2020/9/7	5, 6
32	2020/1/23	5, 6	42	2020/5/22	5, 6	52	2020/9/19	5, 6
33	2020/2/4	5, 6	43	2020/6/3	5, 6	53	2020/10/1	5, 6
34	2020/2/16	5, 6	44	2020/6/15	5, 6	54	2020/10/13	5, 6
35	2020/2/28	5, 6	45	2020/6/27	5, 6	55	2020/10/25	5, 6
36	2020/3/11	5, 6	46	2020/7/9	5, 6	56	2020/11/6	5, 6
37	2020/3/23	5, 6	47	2020/7/21	5, 6	57	2020/11/18	5, 6
38	2020/4/4	5, 6	48	2020/8/2	5, 6	58	2020/11/30	5, 6
39	2020/4/16	5, 6	49	2020/8/14	5, 6	59	2020/12/12	5, 6
40	2020/4/28	5, 6	50	2020/8/26	5, 6	60	2020/12/24	5, 6
Year: 2020, Relative Orbit: 84								
No.	**Acquisition Date**	**Slice**	**No.**	**Acquisition Date**	**Slice**	**No.**	**Acquisition Date**	**Slice**
61	2020/1/1	1	71	2020/4/30	1	81	2020/8/28	1
62	2020/1/13	1	72	2020/5/12	1	82	2020/9/21	1
63	2020/1/25	1	73	2020/5/24	1	83	2020/10/3	1
64	2020/2/6	1	74	2020/6/5	1	84	2020/10/15	1
65	2020/2/18	1	75	2020/6/17	1	85	2020/10/27	1
66	2020/3/1	1	76	2020/6/29	1	86	2020/11/8	1
67	2020/3/13	1	77	2020/7/11	1	87	2020/11/20	1
68	2020/3/25	1	78	2020/7/23	1	88	2020/12/2	1
69	2020/4/6	1	79	2020/8/4	1	89	2020/12/14	1
70	2020/4/18	1	80	2020/8/16	1	90	2020/12/26	1

Table A1. *Cont.*

Year: 2023, Relative Orbit: 157								
No.	**Acquisition Date**	**Slice**	**No.**	**Acquisition Date**	**Slice**	**No.**	**Acquisition Date**	**Slice**
1	2023/1/2	1, 2	11	2023/5/2	1, 2	21	2023/8/30	1, 2
2	2023/1/14	1, 2	12	2023/5/14	1, 2	22	2023/9/11	1, 2
3	2023/1/26	1, 2	13	2023/5/26	1, 2	23	2023/9/23	1, 2
4	2023/2/7	1, 2	14	2023/6/7	1, 2	24	2023/10/5	1, 2
5	2023/2/19	1, 2	15	2023/6/19	1	25	2023/10/17	1, 2
6	2023/3/3	1, 2	16	2023/7/1	1	26	2023/11/10	1, 2
7	2023/3/15	1, 2	17	2023/7/13	1	27	2023/11/22	1, 2
8	2023/3/27	1, 2	18	2023/7/25	1	28	2023/12/4	1, 2
9	2023/4/8	1, 2	19	2023/8/6	1	29	2023/12/16	1, 2
10	2023/4/20	1, 2	20	2023/8/18	1, 2	30	2023/12/28	1, 2
Year: 2023, Relative Orbit: 55								
No.	**Acquisition Date**	**Slice**	**No.**	**Acquisition Date**	**Slice**	**No.**	**Acquisition Date**	**Slice**
31	2023/1/7	5, 6	41	2023/5/7	5, 6	46	2023/9/4	5, 6
32	2023/1/19	5, 6	42	2023/5/19	5, 6	47	2023/9/16	5, 6
33	2023/1/31	5, 6	43	2023/5/31	5, 6	48	2023/9/28	5, 6
34	2023/2/12	5, 6			5, 6	49	2023/10/10	5, 6
35	2023/2/24	5, 6			5, 6	50	2023/10/22	5, 6
36	2023/3/8	5, 6	44	2023/7/6	5, 6	51	2023/11/3	5, 6
37	2023/3/20	5, 6			5, 6	52	2023/11/15	5, 6
38	2023/4/1	5, 6			5, 6	53	2023/11/27	5, 6
39	2023/4/13	5, 6			5, 6	54	2023/12/9	5, 6
40	2023/4/25	5, 6	45	2023/8/23	5, 6	55	2023/12/21	5, 6
Year: 2023, Relative Orbit: 84								
No.	**Acquisition Date**	**Slice**	**No.**	**Acquisition Date**	**Slice**	**No.**	**Acquisition Date**	**Slice**
56	2023/1/9	1	65	2023/5/9	1	75	2023/9/6	1
57	2023/1/21	1	66	2023/5/21	1	76	2023/9/18	1
		1	67	2023/6/2	1	77	2023/9/30	1
58	2023/2/14	1	68	2023/6/14	1	78	2023/10/12	1
59	2023/2/26	1	69	2023/6/26	1	79	2023/10/24	1
60	2023/3/10	1	70	2023/7/8	1	80	2023/11/5	1
61	2023/3/22	1	71	2023/7/20	1	81	2023/11/17	1
62	2023/4/3	1	72	2023/8/1	1	82	2023/11/29	1
63	2023/4/15	1	73	2023/8/13	1	83	2023/12/11	1
64	2023/4/27	1	74	2023/8/25	1	84	2023/12/23	1

Table A2. The parameters for RiceLSTM, RiceMU, and RiceTS models.

Model	**Input Size**	**Reshaped Size**	**Ouput Size**	**Loss Function**	**Total Parameters**	**Estimated Total Size (MB)**
RiceLSTM	256 × 256 × 15 × 2	65,536 × 15 × 2	256 × 256 × 2	CrossEntropyLoss	134,402	1016.08
RiceLSTMWith Attention	256 × 256 × 15 × 2	65,536 × 15 × 2	256 × 256 × 2	CrossEntropyLoss	151,042	2022.78
RiceMU	256 × 256 × 15 × 2	256 × 256 × 30	256 × 256 × 2	CrossEntropyLoss	2,964,234	840.36
RiceTS	256 × 256 × 15 × 2	65,536 × 15 × 2	256 × 256 × 2	CrossEntropyLoss	3,133,066	4520.43

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